

Lakshmana Rao Vallapuneni

Data Engineer | Azure Data Engineer

Grand Rapids, MI | (616) 329-4914 | vlakshmanarao05@gmail.com | [linkedin.com/in/lakshmanarao-vallapuneni/](https://www.linkedin.com/in/lakshmanarao-vallapuneni/)

SUMMARY

Data Engineer with 4+ years of experience designing and optimizing scalable batch and real-time data pipelines across AWS and Azure cloud platforms. Experienced in building high-performance ETL/ELT solutions using Python, SQL, PySpark, Spark, Databricks, Azure Data Factory, AWS Glue, and Apache Airflow to process millions of records for enterprise analytics. Strong expertise in cloud data warehousing (Snowflake, Redshift, Synapse), Delta Lake, Kafka, dimensional data modeling, data quality, performance tuning, CI/CD, and secure data governance supporting analytics and business intelligence initiatives.

TECHNICAL SKILLS

Programming & Scripting: Python, SQL, PySpark, Spark SQL, Shell Scripting

Big Data & Streaming: Apache Spark, Databricks, Hadoop, Kafka, Spark Structured Streaming, Delta Lake

ETL/ELT & Orchestration: Azure Data Factory, AWS Glue, Apache Airflow, dbt, SSIS

Cloud Platforms: AWS (S3, Redshift, Glue, EMR, Lambda, Athena), Azure (Databricks, ADLS Gen2, Synapse, Azure DevOps)

Data Warehousing: Snowflake, Amazon Redshift, Azure Synapse, Dimensional Modeling (Star & Snowflake Schemas), Medallion Architecture

Databases: SQL Server, PostgreSQL, MySQL, T-SQL, Query Optimization, Indexing

BI & Visualization: Qlik, Power BI, Tableau

DevOps & Version Control: CI/CD, Azure DevOps, Git, GitHub, Docker

Methodologies: Agile, Scrum, Data Pipeline Optimization, Performance Tuning

PROFESSIONAL EXPERIENCE

Deloitte, Grand Rapids, MI

July 2025 – Present

Data Engineer

- Engineered scalable ETL/ELT pipelines using SQL, Python, PySpark, and AWS (Glue, S3, Redshift, Lambda) to power enterprise reporting, analytics, and Qlik dashboards that processed millions of records daily.
- Orchestrated and scheduled workflows using Apache Airflow, building dependency-aware DAGs with automated retries, alerting, and SLA monitoring to ensure reliable pipeline execution.
- Optimized SQL reporting queries via execution-plan analysis, join refactoring, and indexing strategies, reducing runtime from 120 seconds to 3 seconds.
- Built automated data quality and validation frameworks to test PySpark outputs, SQL logic, and source-to-target mappings, ensuring accuracy and integrity across reporting pipelines.
- Handled regulated datasets containing PII and PHI, enforcing secure access controls, data masking, and auditability in compliance with governance standards.

Tech Mahindra, India

Apr 2022 – Jul 2024

Azure Data Engineer

- Drove enterprise migration of legacy on-prem ETL workloads to Azure (Data Factory, Databricks, ADLS Gen2, Azure SQL DW, Spark), retiring brittle legacy jobs and cutting infrastructure and maintenance overhead.
- Implemented Medallion Architecture (Bronze, Silver, Gold) in ADLS Gen2, improving data organization, lineage, and reusability across analytics layers.
- Developed high-volume data ingestion and transformation workflows using PySpark and Spark SQL, optimizing performance through partitioning, caching, and broadcast joins.
- Monitored and supported production pipelines post-migration, diagnosing job failures and data discrepancies to maintain SLA compliance for downstream telecom reporting.
- Established Azure DevOps CI/CD pipelines, linked repositories to the development environment, and automated code change deployment across development, UAT, and production environments.

HCLTech, India

May 2021 – Mar 2022

Associate Data Engineer

- Developed batch and real-time data ingestion frameworks using Apache Kafka, Spark Structured Streaming, REST APIs, and CDC processes to support low-latency processing of high-volume enterprise data.
- Designed and developed ETL workflows using Azure Data Factory, Azure Databricks, and PySpark to automate data ingestion, cleansing, transformation, and loading into cloud-based analytical environments.
- Modeled and implemented cloud data warehouse solutions in Snowflake using Star and Snowflake schemas, enabling efficient querying and self-service reporting for business users.
- Implemented automated source-to-target reconciliation and data validation processes to improve data quality, reduce manual verification efforts, and ensure consistency across downstream reporting systems.

EDUCATION

Master of Science in Data Science and Analytics, Grand Valley State University, Grand Rapids, MI

Apr 2026

CERTIFICATIONS

- AWS Certified Machine Learning
- Microsoft Certified: Azure Data Engineer Associate